Problem 2 - Gallery Description Cards

Week 7 | Category B Assignment | Java solutions

Problem

Create an abstract ArtPiece with a unique final pieceId generated by a shared static counter. Painting and Sculpture extend it directly.

Complete Java Code

```
abstract class ArtPiece {
    private static int nextId = 1001;
    private final String pieceId;

    protected ArtPiece() {
        pieceId = "ART-" + nextId++;
    }

    public abstract String describe();

    public String getPieceId() {
        return pieceId;
    }
}

class Painting extends ArtPiece {
    private final String title;

    public Painting(String title) {
        super();
        this.title = title;
    }

    @Override
    public String describe() {
        return "Painting: " + title + ", framed on canvas";
    }
}

class Sculpture extends ArtPiece {
    private final String title;

    public Sculpture(String title) {
        super();
        this.title = title;
    }

    @Override
    public String describe() {
        return "Sculpture: " + title + ", carved from stone";
    }
}

public class Main {
    public static void main(String[] args) {
        Painting p = new Painting("Sunset Fields");
        Sculpture s = new Sculpture("The Thinker II");
        System.out.println(p.describe());
        System.out.println(s.describe());
        System.out.println(p.getPieceId());
        System.out.println(s.getPieceId());
    }
}
```

Sample Output

```
Painting: Sunset Fields, framed on canvas
Sculpture: The Thinker II, carved from stone
ART-1001
ART-1002
```

How it works

The abstract class cannot be instantiated. Its constructor increments the static counter once per ArtPiece object, and pieceId is final after construction. The sample uses ART-1001 and ART-1002 as the chosen ID format.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.